

CafeTrack: Real-Time Stock Monitoring with QR-Based Authentication for Merrylane Cafe Foodhub

IT System Project Document

1. Project Overview

Project Title: CafeTrack: Real-Time Stock Monitoring with QR-Based Authentication for Merrylane Cafe Foodhub

Project Description

CafeTrack is a connected inventory management system designed to automate and streamline the management of ingredient stocks, item checkout, restock operations, and expiration tracking for Merrylane Cafe Foodhub. It provides a centralized, secure, and real-time platform for recording stock movements, generating QR codes, calculating inventory levels, producing reports, and ensuring accountability through an immutable audit trail.

The system reduces manual effort, minimizes human error, and enhances transparency for both staff and administrators. Through QR-based authentication and real-time stock deduction, CafeTrack enables the cafe to maintain accurate inventory records while preventing spoilage, waste, and stock discrepancies.

Group Members:

- **Project Lead / Backend Developer:** Maquinto, Mark Jhaeyv
- **System Analyst / Frontend Developer:** Obedencio, Mark Andrei
- **System and Logic Tester / Documentation:** Rodriguez, Prince Deano

Client Group: Merrylane Cafe Foodhub, Lipa City

Developer Group: Group 3

Date of Agreement: November 16, 2025

Document Version: 1.0

2. Project Goals and Objectives

Primary Objective

To develop a connected inventory management system that allows administrators to manually encode and generate QR codes for ingredients, while enabling staff to perform quick, QR-based checkout and real-time stock deduction for Merrylane Cafe Foodhub.

Additional Objectives

Beyond the core client requirements, the system includes the following strategic enhancements to improve usability, accountability, and operational effectiveness:

- **Role-Based Access Control (RBAC):** Distinct permissions for Admins (owner/manager) who manage the system, and Staff (cooks/baristas) who can only perform checkout and restock operations.
- **QR Code/Barcode Authentication (Staff Login):** Staff scan their personal QR/barcode codes to log in and enter a password to verify their identity before performing any inventory transactions.
- **Manual Item Management (Admin Check-In):** Admins can manually add, edit, or delete ingredients through the web dashboard, inputting details such as category, physical form (liquid/powder), quantity, and expiration dates.
- **Automatic QR Code Generation:** The system automatically generates unique QR codes or barcodes for each ingredient upon manual entry, which can be printed and affixed to storage containers.
- **QR-Based Item Checkout/Restock:** Staff can scan item QR codes using connected scanner devices to quickly and accurately deduct items from inventory.
- **Multi-Item Checkout with Confirmation:** Staff can scan multiple items in one transaction, with a final confirmation step (password/code) required to prevent accidental deductions.
- **Real-Time Stock Updates:** Inventory levels automatically adjust instantly in the database and reflect on the admin dashboard immediately after every checkout transaction.
- **Automated Alerts & Reports:** The system generates real-time alerts for low-stock levels and near-expiry items, plus historical reports for inventory usage and waste tracking.
- **Immutable Audit Trail:** A secure, uneditable chronological log records all system activities (item additions, checkouts, adjustments) with staff identification and timestamps for compliance and accountability.
- **Expiration & Out-of-Stock Alert:** The system alerts admin users about items that are near expiration, out of stock, or already expired.

3. Requirements

Client Requirements

Functional Requirements

User Authentication

- Secure login for Admin and Staff users.
- Staff authentication via personal QR code scan plus password verification.
- Admin authentication via username and password.
- Session management with automatic timeout after inactivity.
- Failed login attempts logged and monitored.

User Management (Admin)

- Add, edit, disable, and enable user accounts.
- Assign roles: Admin (owner/manager) and Staff (cooks/baristas).

- Generate and assign personal QR/barcode credentials per staff member.
- View all users with search and filtering capabilities.
- System prevents disabling the last active Admin account.

Item Management (Admin)

- Add, edit, and delete ingredients through the web dashboard.
- Input item details: category, physical form (liquid/powder), quantity, and expiration date.
- Generate unique QR codes or barcodes automatically upon item entry.
- Print generated QR codes for physical labeling and attachment to storage containers.
- View all inventory items with search, filter, and sort capabilities.
- Track item quantities in real time with automatic deduction upon checkout.

QR-Based Inventory Checkout/Restock

- Staff must authenticate via personal QR code and password before performing any transaction.
- Staff can scan item QR codes using connected scanner devices.
- System automatically deducts scanned items from inventory during checkout.
- Support multi-item checkout within a single transaction.
- Final confirmation step (password/code) required before completing a checkout.
- Restock operations supported with automatic quantity addition.
- Invalid or unrecognized QR codes are rejected with error messages.
- Manual entry of item ID available as fallback for damaged QR codes.

Real-Time Stock Monitoring

- Inventory levels update instantly in the database after every transaction.
- Admin dashboard reflects current stock levels in real time.
- Real-time alerts triggered for low-stock levels and near-expiry items.
- Separate alerts for out-of-stock and expired items.
- Stock deduction uses optimistic locking to prevent negative inventory.
- Concurrent checkouts are handled without overselling.

Reporting & Analytics

- Generate historical reports for inventory usage and waste tracking.
- Filter and sort reports by date range, item category, and staff.
- Export inventory reports for documentation (PDF/CSV).

Audit Trail

- Log all system activities: item additions, checkouts, adjustments, and restocks.
- Record staff identification and timestamps for every transaction.
- Maintain an immutable, uneditable chronological log for compliance.
- Support audit filtering by staff, date, and transaction type.

- Log failed login attempts and unauthorized scan attempts.
- Hash chaining for tamper detection.

Administrator Functions

- Manage company settings: item categories, storage locations, alert thresholds.
- Configure low-stock thresholds and expiration warning periods.
- View and monitor all system activity through the admin dashboard.
- Resolve alerts after taking corrective action.

Non-Functional Requirements

Usability

- Intuitive interface designed for all user roles (Admin, Staff).
- Simple navigation with a clean, organized dashboard layout.
- Role-based tab navigation showing only relevant modules per user.
- Consistent modern styling with clear prompts and status labels.
- Minimal learning curve for staff performing checkout operations.

Security

- Role-based access control (RBAC) restricting functions per user role.
- Secure login authentication via QR/barcode scan and password for staff.
- Username and password authentication for admin.
- Encrypted storage of sensitive user data and credentials (AES-256).
- Encrypted data transmission (TLS 1.3).
- Session management with automatic timeout after 15 minutes of inactivity.
- Admin protection preventing system-critical action disruption.
- Last active Admin account cannot be disabled.

Performance

- Fast QR scan processing with near-instant stock deduction (under 1 second).
- Responsive dashboard updates with real-time data refresh (under 2 seconds).
- Support for simultaneous scanning and checkout transactions.
- Efficient filtering, sorting, and search operations across large inventories.
- Database reads for item lookup complete in under 100 milliseconds.

Reliability

- Data persistence across sessions with reliable database storage.
- Validation locks on records after finalization.
- Exception handling and error recovery mechanisms.
- Policy-based constraints to enforce system integrity (no negative stock).
- Backup and restore capability for inventory data.

- Target 99.5% uptime during operating hours.

Scalability

- Support for multiple item categories and storage locations.
- Ability to handle thousands of item records efficiently.
- Modular architecture supporting feature expansion.
- Design accommodating additional branches or expanded menu items.

Maintainability

- Modular architecture enabling easy feature updates.
- Service layer pattern for separation of concerns (InventoryService, AuthService, ReportService).
- Configurable system settings for policy adaptation.
- Domain model with clear entity relationships.
- Documented codebase supporting future developers.

4. Design and Development

User Interface (UI) Design

The CafeTrack system features a clean, modern interface designed for both administrative and staff users. The UI follows a dashboard-based layout with role-specific navigation.

Planned Screens

- 1. **Login Page** — QR scan input field with password verification for staff; username/password form for admin.
- 2. **Admin Dashboard** — Overview of current stock levels, low-stock alerts, near-expiry alerts, and recent transactions.
- 3. **Item Management Page** — Add/edit/delete ingredients with fields for category, physical form, quantity, and expiration date; QR code generation and print functionality.
- 4. **Checkout/Restock Page** — QR scanner input, multi-item scan list, and final confirmation button.
- 5. **Reports Page** — Tab-based interface with usage reports, waste tracking, and export options.
- 6. **Audit Trail Page** — Chronological log with filters for staff, date, and transaction type.
- 7. **User Management Page** — Create/edit/disable user accounts, assign roles, and generate QR credentials.
- 8. **Alert Center** — Dashboard showing active low-stock, near-expiry, out-of-stock, and expired item alerts with resolution options.

5. Project Plan

Development Timeline

Week	Planned Task	Status	Planned Completion Date	Actual Completion Date
1	Requirements gathering and analysis; client interview with Merrylane Cafe Foodhub	To be determined	—	—
2	System design: ERD, DFD, UI wireframes; QR code library research	To be determined	—	—
3	Planning system execution; distributing tasks among members; setting up development environment	To be determined	—	—
4	Documentation defense; start of system development (item management, QR generation)	To be determined	—	—
5	System development phase (QR authentication, checkout/restock, real-time updates)	To be determined	—	—

Week	Planned Task	Status	Planned Completion Date	Actual Completion Date
6	Final system completion; developer testing; integration testing	To be determined	—	—
7	Final project presentation and user acceptance testing	To be determined	—	—

Testing and Quality Assurance

Unit Testing (Developer Team)

- QR code generation and validation logic
- Staff authentication (QR scan + password)
- Item checkout and restock deduction logic
- Real-time stock update triggers
- Low-stock and near-expiry alert thresholds
- Audit trail logging and immutability
- Role-based access control enforcement
- Report generation and filtering

Integration Testing (QA Team + Developers)

- End-to-end staff login → scan items → checkout → stock deduction → audit log
- Admin item creation → QR generation → print → staff scan
- Restock workflow → stock update → alert clearance
- Multi-item checkout with confirmation step
- Expiration alert triggering → admin notification
- Low-stock alert triggering → reorder suggestion
- Audit trail retrieval and filtering

System Testing

- Verify all traced requirements (R1–R15) function correctly
- Validate real-world workflows match business processes
- Test usability for all user roles
- Verify compliance with food safety and inventory standards
- Performance testing with realistic item volumes (e.g., 500+ items)
- QR scanning reliability under various conditions (lighting, angle, distance)

User Acceptance Testing (UAT)

- Conducted with Merrylane Cafe Foodhub staff and management
- Confirm all requirements are met

- Validate real-world workflows
- Ensure usability and compliance with business rules

Testing Plan and Results

Category	Test Case	Status	Result
Authentication	Staff QR login with valid credentials	To be tested	—
Authentication	Staff QR login with invalid credentials	To be tested	—
Authentication	Admin username/password login	To be tested	—
Item Management	Add new ingredient with all details	To be tested	—
Item Management	Generate and print QR code	To be tested	—
Item Management	Edit ingredient details	To be tested	—
Item Management	Delete ingredient	To be tested	—
Checkout	Scan single item and deduct stock	To be tested	—
Checkout	Multi-item checkout with confirmation	To be tested	—
Checkout	Invalid QR scan rejection	To be tested	—
Restock	Scan item and add stock	To be tested	—
Real-Time Updates	Dashboard reflects stock changes immediately	To be tested	—
Alerts	Low-stock alert triggered at threshold	To be tested	—
Alerts	Near-expiry alert triggered	To be tested	—
Alerts	Out-of-stock alert triggered	To be tested	—
Audit Trail	All transactions logged with staff ID and timestamp	To be tested	—
Audit Trail	Audit records cannot be modified or deleted	To be tested	—
Reports	Inventory usage report generated	To be tested	—
Reports	Waste tracking report generated	To be tested	—
Security	RBAC enforced across all modules	To be tested	—
Security	Last admin account protection	To be tested	—

6. Implementation Details

Environment Setup

The development of CafeTrack will require a properly configured environment to ensure smooth coding, testing, and execution. The following tools and technologies are planned:

1. **Integrated Development Environment (IDE):** Eclipse IDE or Visual Studio Code, chosen for its user-friendly interface and strong support for Java/web development.
2. **Programming Language:** Java SE (Standard Edition) for the desktop application, or JavaScript/TypeScript for a web-based implementation. The final choice depends on client infrastructure.
3. **GUI Framework:** Java Swing (for desktop) or React/Vue.js (for web), used for building the interface.
4. **Local Storage / Database:** File-based storage using Java I/O (for desktop) or a relational database such as PostgreSQL or MySQL (for web-based implementation).
5. **QR Code Library:** ZXing (Zebra Crossing) or a similar library for generating and decoding QR codes.
6. **Scanner Integration:** USB barcode/QR scanner support through standard input handling.

Libraries and Dependencies

- Java built-in libraries: `java.io`, `java.util`, `javax.swing` (for desktop)
- QR code generation: ZXing library
- Serialization: Java Object Serialization (for file-based persistence)
- Database connector: JDBC (if using a relational database)
- Web framework: Spring Boot or Express.js (if web-based)

7. Reflection on Sustainable Development Goals (SDGs)

SDG Alignment Summary

The development of CafeTrack supports several United Nations Sustainable Development Goals by promoting efficiency, sustainability, and responsible management practices within the food service industry. Although it is a digital system, its features contribute to broader social, economic, and environmental goals.

SDG 8 — Decent Work and Economic Growth

CafeTrack makes cafe tasks easier, helping staff work faster and with fewer mistakes. This leads to better service, smoother operations, and improved business growth. By automating inventory tracking and reducing manual errors, the system supports decent work conditions through:

- Reduced physical and mental strain on staff
- Transparent and fair inventory accountability
- Improved operational efficiency and business sustainability
- Enhanced workplace transparency and trust

SDG 9 — Industry, Innovation, and Infrastructure

CafeTrack modernizes cafe operations by digitalizing inventory management, QR-based checkout, and real-time stock monitoring. This reduces errors, makes operations more efficient, and strengthens the cafe's digital infrastructure through:

- Adoption of QR technology for authentication and tracking
- Real-time data synchronization and dashboard analytics
- Automated alerts and reporting for proactive decision-making
- Scalable architecture supporting future expansion

Impact and Future Enhancements

In the future, CafeTrack can be improved to create a bigger impact on the selected SDGs. Possible enhancements include:

- AI-powered demand forecasting to further reduce waste
- Integration with POS systems for automated sales-based inventory deduction
- Mobile application for remote monitoring and alerts
- Multi-branch support with centralized dashboard
- Supplier management and automated reordering
- Sustainability metrics and eco-friendly reporting

These improvements will make operations more efficient, support sustainable growth, and further modernize cafe services.

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